Lab7 Requirements

1. 新增 system call
   - Raspberrypi3_defconfig (32-bit Linux)

     確認在 Syscall.tbl 增加 mysys 編號與 test.c 相同

```
jie@lab304-jie:~/Desktop/embedding_system/raspberrypi3_32/buildroot/output/build/linux-custom/arch/arm/tools$ tail syscall.tbl
442     common  mount_setattr                  sys_mount_setattr
443     common  quotactl_fd                    sys_quotactl_fd
444     common  landlock_create_ruleset        sys_landlock_create_ruleset
445     common  landlock_add_rule              sys_landlock_add_rule
446     common  landlock_restrict_self         sys_landlock_restrict_self
# 447 reserved for memfd_secret
448     common  process_mrelease               sys_process_mrelease
449     common  futex_waitv                    sys_futex_waitv
450     common  set_mempolicy_home_node        sys_set_mempolicy_home_node
451     common  mysys                          sys_mysys
```

   - Raspberrypi3_64_defconfig (64-bit Linux) (可參考 lab7 投影片)

     確認在 unistd.h 增加 mysys 編號與 test.c 相同

```
jie@lab304-jie:~/Desktop/embedding_system/raspi3_64/buildroot/output/build/linux-custom$ cat include/uapi/asm-
generic/unistd.h | grep 451
#define __NR_mysys 451
#define __NR_syscalls 452 //451
```

2. 編譯 test.c

```c
// test.c
#include <linux/kernel.h>
#include <sys/syscall.h>
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <errno.h>

#define __NR_mysyscall 451


long mysyscall_syscall(void)
{
    return syscall(__NR_mysyscall);
}

int main(int argc, char *argv[])
{
    long activity;
    activity = mysyscall_syscall();
"test.c" 30L, 579B                                                    1,1           Top
```

3. 到樹梅派上執行 text.exe

```
Welcome to Buildroot
jie login: root
# ../test_64.exe
[   35.628156] Hello System Call.
Congratulations! Your system call is functional. Run the command dmesg in the terminal and find out!
#
```

4. 用 dmesg 確認

```
[    6.073392] lan78xx 1-1.1.1:1.0 eth0: Link is Down
[   35.628156] Hello System Call.
#
```

DEMO 時直接在樹梅派上執行 test.exe 並用 dmesg 確認即可。